

The emergence of “weak” definite articles¹

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Abstract. This paper investigates the emergence and distribution of weak and strong definite articles in German. While reduced weak forms are already attested in Old High German, their use was initially rare and largely restricted to noun phrases that are very likely to have a singleton extension in any situation. Over time, weak articles expanded to a wider range of noun phrases, yet systematically failed to extend to anaphoric environments with an indefinite antecedent, where strong forms remain obligatory in Modern German. We aim to explain this diachronic trajectory by combining evidence from historical corpora with models from computational pragmatics and agent-based simulations using reinforcement learning.

Keywords: Weak definite articles, computational pragmatics, uniqueness

1. Introduction

Since Heinrichs (1954) two morpho-phonological classes of definite articles have been distinguished in a number of Germanic varieties: “weak” and “strong” ones. The distinction has been most thoroughly documented and discussed, in particular, for Standard German and for Fering Frisian (Schwarz 2013 for references). Weak definite articles are also attested in Middle Dutch as nominal proclitics, they are absent from the Modern Dutch (*tkint* ‘the child’, from Corpus Gyseling (2021) w.997745, cf. Present Day Dutch *het kind*). In German the “weak” paradigm has the status of prepositional enclitics: for example, in *vom* (‘of the’) vs. the “strong” one *von dem* (‘of the’). According to Waldenberger (2009), they are attested already in Old High German, but Coniglio and Schlachter (2014) point out that the use of reduced weak articles was quite rare at that stage and was mostly constrained to NPs whose referents “have to be unique” (e.g. *zur helle* ‘to the Hell’ or *ans cruce* ‘to the Cross’). In Modern German (and Fering Frisian), “weak” forms cover all the contexts covered by the English *the* with the exception, importantly, of the context involving an indefinite antecedent in the preceding sentence, where a “strong” form is used (Ebert 1971, Schwarz 2009). Our goal is to understand *why* the trajectory of the reduced forms is the way it is, namely, spreading from NPs necessarily having unique referents to other types before stopping at the anaphoric “watershed”. We set out to model the historical emergence of the morphophonological and distributional distinctions leveraging historical corpora, computational pragmatics, and reinforcement learning agent-based simulations. The core intuition we pursue is that weak articles can give rise to referential uncertainty, which initially constraints them to NPs where the risk of reference error is minimal, and that with further use the error tolerance increases, contributing to their spread across lexical items.

2. Preliminary corpus Study

Before we propose diachronic mechanisms for the emergence of “weak” definite articles and test them using computational simulations, we first examine the phenomenon as it unfolds in its

¹We thank the audience at Sinn und Bedeutung for helpful comments and discussion.

natural historical development, and then model the transition by making explicit assumptions about the semantics of the forms and the mechanisms driving diachronic change.

To reach this goal, we performed a corpus query on the Indiana Parsed Corpus of Historical (High) German (IPCHG) (Sapp et al., 2024), with 55 texts ranging from 1250 to 1650. Figure 1 shows the relative frequency of the different D-heads in PPs: a proximal or distal demonstrative, a weak or strong article, a weak feminine form (such as *beir*, which is not productive in Modern German) or a bare noun. The data have been binned into 50-year intervals.

While bare nouns are the most diffused category, weak articles increase at the expenses of strong ones, particularly around 1500. This confirms that weak articles spread at a later stage in the history of German. Regarding the contexts of emergence, our results are in line with Coniglio and Schlachter (2014): the earliest attestations of weak articles are found with NPs whose extensions are singletons in any situation where they do have an extension (cf. Hawkins 1978’s global situation uses of definite articles). Some examples of such attestations are *im paradis* (‘in paradise’), *im himel* (‘in the sky’), *zunc heiligen Crucze* (‘to the Holy Cross’), *zur tauelrunden* (‘to the Round Table’). Some full examples are reported in (1) and (2).

- (1) *Do der kuonik der eren vnser here ihenc xpc zvr helle qam [...]*
when the king DEF.GEN glory our lord Jesus Christ to.DEF.W hell came

“When the King of Glory, Our lord Jesus, went to hell [...]”

(*Leipziger Rel-Sermon ThurSax.* 77, 1300)

- (2) *Sie sach ein swester die waz urschaiden auz irm closter im paradis.*
she saw a sister who was separated from her monastery in.DEF.W paradise

“She saw a sister who was away from the monastery, in paradise.”

(*Engelthaler Rel-Narrative* 568, 1340)

Importantly, at this stage, unlike in Modern German, strong forms can also be used with such NPs. 3 is taken from the same passage as 1 but contains a full article form with the NP *helle*.

- (3) *Do der kuonik der eren, vnser ihenc xpc, zvr helle qam, vnd in do*
when the king DEF.GEN glory our Lord Jesus to hell came and him then
gesahen die heiligen lute, die da zv der helle warn.
saw DEF holy people that then in DEF.S hell were

“When the King of Glory, our Lord Jesus, went to hell, then the holy folk who were there in hell saw him.”

(*Leipziger Rel-Sermon ThurSax.* 77, 1300)

The earliest attestations of weak articles also include NPs denoting functional relations e.g. *am mittewochen vor michael* (‘on the Wednesday before Michaelmas’) or *zum dritten mal* (‘on the third time’), which, again, have a singleton extension wherever they have an extension. Occasionally, we find reduced articles with NPs for which the probability of a singleton extension in a random situation where they have an extension is, intuitively, smaller yet still appreciable, e.g. *vom Couent* (‘of the convent’) (cf. Hawkins 1978’s larger situation uses of definite articles).

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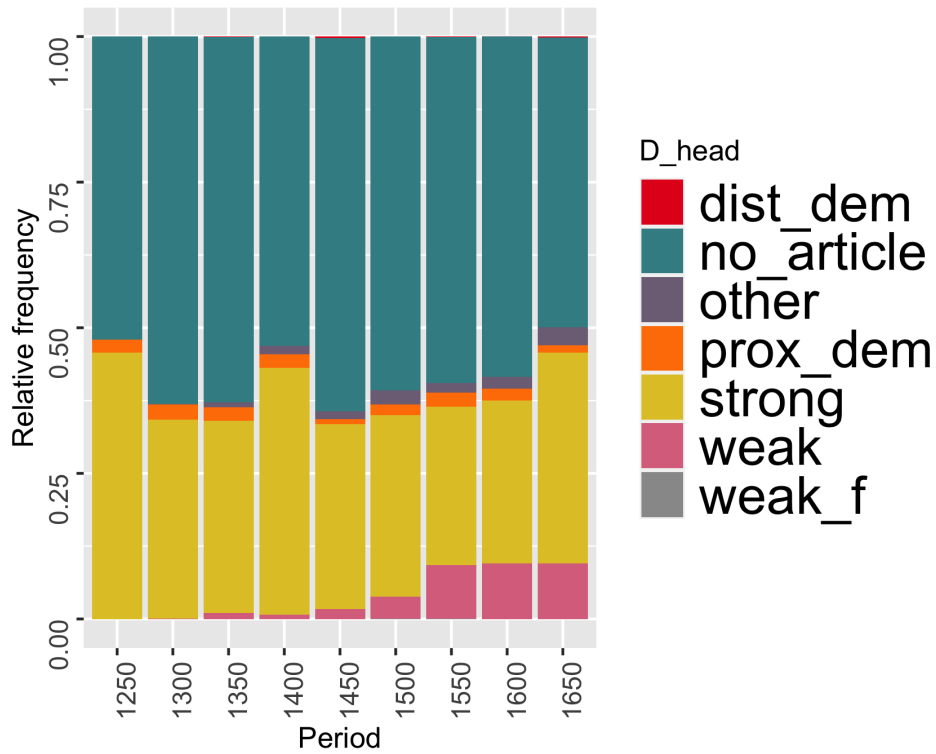


Figure 1: The distribution of different D-exponents in High German Prepositional PPs in the IPCHG corpus.

Later in the corpus we observe a progressive diffusion of the weak articles to NPs which can be associated with a yet lower probability of having a singleton extension in a randomly taken situation e.g. *zur herberge* (‘to the inn’), *zur stat* (‘to the city’), *zum karch* (‘to the church’), *zum bette* (‘to bed’), *zur brucken* (‘to the bridge’), indicating a relaxation of the initial constraint.

- (4) *Da sagt yn myn herre Gawan wie er **zum** **karch** komen was [...]*
 Then said him my lord Gawan how he to.DEF.W church come was
 “Then my lord Gawan told him how he had come to the church [...]” (*Karrenritter Lit-prose* 207, 1430)
- (5) *Er qwam wiedder **zum** **bette** vnd die Jungfrau fraget yn was er thun wolt*
 He came back to.DEF.W bed and the maiden asked him what he do wanted
 “He returned to the bed and the maiden asked him what he wanted to do” (*Karrenritter Lit-prose* 239, 1430)

Hence, our data confirm Coniglio and Schlachter (2014)’s observation on the emergence of reduced articles, which started from nouns with a high likelihood of having a singleton extension to a more progressive relaxation of this constraint. In the next sections, we set out to reproduce these results with a computational simulation, which is based on a theoretical model which combines situation semantics, game-theoretic setups, and hypotheses on phonetic reduction.

3. Model

We propose that the empirically observed, diachronically dynamic, relation between the article choice and NP classes is a result of an interplay between (constant) semantico-syntactic representations of articles (constant) pragmatic principles of economy and clarity, and (constant) speakers' tendency to incrementally increase economy if it has been proven not to compromise clarity *too much*, specifically, as a function of communicative successes. To test our model, we use a game-theoretic setup, augmented with a reinforcement learning algorithm.

The semantic part of our model is based on the intuition that weak DPs are always interpreted with respect to a topic situation, where the relevant NP may or may not have a singleton extension, depending on the topic situation, the Common Ground, and the lexical type of the NP. Strong article-NPs, in contrast, are interpreted with respect to a contextually salient situation in which the NP is presupposed to have a singleton extension.

We take, by now standard, Schwarz (2009)'s semantics of weak articles in 6 as a starting point.

$$(6) \quad \llbracket D \rrbracket = \lambda s_s . \lambda P_{\langle e \langle s, t \rangle \rangle} : \exists ! x [P(x)(s)] . \iota x [P(x)(s)]$$

Pace Schwarz (2009), who assumes that both weak and strong articles are associated with an LF which involves a situation pronoun adjoined to D, which can be left free or bound by a higher binder, we assume that weak articles are exponents of a D without a situation pronoun.² The situation argument of such DP has to be valued, eventually, by the topic situation. The latter, we assume, following Schwarz (2009: 87), is syntactically represented at the left edge of the clause. Whether such a (single) DP refers depends, naturally, on whether the topic situation contains a unique individual with the nominal property. For instance, in 5, the Question Under Discussion, from which Schwarz (2009) proposes to derive the topic situation, is presumably about where the person in question (Lancelot) return (after he got up from the bed he was lying on in a chamber in a splendid castle he wandered into). Taking the topic situation to be the sum of minimal situations where the actual world answers to the QUD hold (Schwarz (2009: 127-133) expanding on Kratzer (2021) who makes use of a situation-based version of Groenendijk and Stokhof (1984)'s semantics of questions), the topic situation in 5 is the sum of minimal situations where Lancelot returned to the same place as he did in the “actual” Lancelot's world in the chamber he had entered after getting up from the bed he was lying on. This topic situation involves just one bed because it's the minimal situation of returning to the unique place (of departure). If instead of *qwam wiedder* it said “walked to” and therefore the topic situation involved an event of Lancelot walking in the chamber, *zum bette* would fail to refer since, as the story goes, the chamber was equipped with two beautiful beds. If, on the other hand, “walked to” was used but nothing were said in the previous context about the number of available beds, the hearer of 5 would likely imagine a chamber with a single bed (a typical chamber in a medieval castle), and the communication would succeed. It follows from the QUD-based approach to topic that there is a degree of uncertainty on the part of the hearer about the exact contents of the topic situation. The success or failure of communication in part depend on how the hearer decides to “reconstruct” the missing details, such as the number of beds in a castle chamber if that is not contextually specified. The speaker, in turn, forms an utterance based on their estimate

²Schwarz (2009: 93) entertains an LF without a situation pronoun for a situation-semantics adaptation of Musan's semantics for cardinal NPs.

about most likely reconstruction scenarios. Whether a given definite DP refers and therefore the success or failure of communication thus depends both on the properties of the topic situation (derived from the QUD) *and* on the likelihood of a particular nominal lexeme, in this case a lexeme meaning “bed”, to have a singleton extension in a random situation. Crucially, we assume that the speakers take into account both parameters when choosing a particular morphological form to convey their message.

We operationalize the context-independent probability of an NP to have a singleton extension in a random situation as a parameter δ : e.g., an NP meaning ‘paradise’ will have a higher δ than that meaning ‘church’, and the latter will have a higher δ than ‘bed’, since paradise is more likely to be unique in a random situation (with a paradise) than a church, and the latter is more likely to be unique than a bed, and that one is, intuitively, more likely to be unique than an apple.³

We assume that strong articles are semantically exactly like the weak ones (i.e. 6), but that their LF does involve a situation pronoun adjoined to D. That is, strong forms are exponents of the same D as the weak ones, plus a situational component. The variable over situations that the pronoun denotes can be mapped by an assignment function onto a salient value, just as it is the case with personal pronouns. Because of the presupposition introduced by D, the salient value has to be of a particular kind, namely, it should contain a singleton with the nominal property.

- (7) $\llbracket s_i D \text{ dog} \rrbracket^g$ is defined iff there is a unique dog in the situation $g(i)$, if defined, $\llbracket s_i D \text{ dog} \rrbracket^g$ denotes the unique dog in $g(i)$.

Informally, the presence of the pronominal component at the LF of the strong article “urges” the hearer to look for a suitable situation, specifically, a salient situation with a single individual with the NP-property. A situation introduced in the immediately preceding discourse is one possible salient value, which gives rise to anaphoric readings, as suggested in Simonenko and Carlier (2022) and Krifka and Modarresi (2023: 350). The topic situation is another possible choice. Semantically, nothing in our proposal prevents the value of the situation pronoun in the LF of a strong article from coinciding with the topic situation. Pragmatically, however, the use of a morphophonologically heavier strong form when a clitic one is available and can generate the same meaning goes against the economy of expression.

When making an article choice, the speaker thus weighs the communicative risk associated with the clitic form that a given NP will not have a singleton extension in a given topic situation *as construed by the hearer* against the morphophonological heaviness of the strong form. Intuitively, the choice is at least partially determined with the risks associated with a given NP.

We model this in a game-theoretic framework, where the strong NPs are invariably associated with *a situation where the NP has a singleton extension* (their “meaning” in the model) whereas clitic or weak NPs are “ambiguous” in that they can either be associated *either with a situation where the NP has a singleton extension or with a situation where the NP has an extension greater than a singleton*. The hearer picking the latter situation amounts to reference failure.

From this perspective, a fully-fledged phonological realization (i.e. “strong” article form) ensures a higher chance that the hearer will pick the intended referent, leading to communicative success. However, the general preference of speakers to use *hypo-articulated* forms requiring

³This scalar differentiation is reminiscent of the categorical distinction between inherent and accidental uniqueness in Šimík (2018), with NPs with highest δ ’s corresponding to inherently uniquely-denoting ones and the NP with all lower δ ’s to accidentally uniquely-denoting ones.

less articulatory effort (Lindblom, 1990) may lead them to pick the reduced form, which results in a tension between clarity vs. economy in the article choice. These two forces create a trade-off whereby NPs with a higher δ are expected to be more likely to trigger article reduction: if a given NP is very likely to have a singleton extension (e.g., ‘paradise’), the speaker is also more likely to reduce their articulatory effort since they are relatively sure of communicative success. To model the diachronic spread of the reduced forms, we assume that, “emboldened” by each communicative success, the speaker increase their preference for economy with time.

4. Methodology

We tested our model by computationally simulating the evolution and distribution of weak forms based on the assumptions we laid out above. The simulations consisted of a series of referential games with two rational agents per game. The Bayesian reasoning of each agent was modeled using the Python library *Pyro* for probabilistic programming (Bingham et al., 2019). Each agent develops through time an optimal pattern of article use through reinforcement learning implemented in object-oriented programming (OOP). Each agent is an object initially endowed with a series of attributes: a randomly assigned age ≤ 80 , a randomly assigned age of death, a rationality parameter α (1 or 2, randomly assigned) which controls the agent’s ambiguity aversion, and costs penalizing redundant forms (to model economicity) and novel forms (to model conservativity). The job of an agent-speaker is to convey the message assigned to them (“information state”) in the context of a particular NP using either a weak or a strong article form (“utterance”). Our information states are situations where the NP in question has a singleton or greater-than-singleton extension. The job of an agent-listener is, upon hearing the utterance, to decide which information state was meant. Only if the utterance is associated by the listener to a situation containing a singleton extension, is the reference resolution for a definite expression successful. In the lexicon of the game, the forms are associated with one or more information states (ambiguous items). We encode our proposal that weak articles can give rise to a failed reference in case the topic situation does not contain a singleton extension of the relevant NP by “lexically” associating weak articles with both information states.

More technically, in the simulation, 20 couples of agents were involved with two types of information states: $s_1 = \{e_1 \dots e_n\}$ and $s_2 = \{e_1\}$, randomly assigned to a speaker. The states correspond to a situation containing multiple entities with (the relevant) NP-property or a single one, respectively.

The likelihood of a given NP to have a singleton extension, and therefore being associated with $s_2 = \{e_1\}$, is modeled via a commonly assumed prior bias δ , ranging from 0 to 1, in increasing order of likeliness of satisfying $s_2 = \{e_1\}$.

Furthermore, the referential game can occur in anaphoric or non-anaphoric contexts. Anaphoric contexts, that is, context involving an indefinite antecedent in the preceding clause, are a special case with regard to the possibility of satisfying the uniqueness condition of definites. Intuitively, the appearance of an indefinite NP signals that no salient situation is entailed by the common knowledge to contain a singleton with the NP property. That is, the chances that a random topic situation in this context satisfies the singleton condition go down compared to the general estimate. Therefore, in our model, the appearance of an indefinite, “anaphoric context”, has a negative effect on the δ parameter. Specifically, we assume that in such cases the normal δ is replaced by δ' , estimated as in 8, where β is the inverse of δ ($\beta = 1 - \delta$), corresponding to the likelihood of an NP *not* to have a singleton extension in a situation where it has an extension, and

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γ , set to 1.9, is a penalizing parameter, which increases the likelihood of not having a singleton extension.

$$(8) \quad \delta' = 1 - \beta * \gamma$$

δ' is the likelihood that an NP has a singleton extension in a random situation where it has an extension in the presence of an indefinite antecedent.⁴

The speaker, given the randomly assigned state, must choose between two phonetic realizations (u) of the definite article – a fully-fledged form u_1 and a reduced one u_2 – to guide the hearer towards the correct inference of the state. These actions are modeled according to Frank and Goodman (2012)’s framework: each participant is a rational Bayesian agent who will simulate the behaviour of the other interlocutor and will act upon probabilistic reasoning. The probabilistic reasoning and consequent actions are practically implemented with Pyro methods which take priors and utterance/state as input and return a posterior probability distribution of the state/utterance (depending on the role of the agent in each game). Upon this probabilistic reasoning, each agent chooses their action using the *random.choice* method in NumPy: this choice is not completely random, but is constrained by the posterior distribution computed. Specifically, the speaker receives the randomly assigned state to express and the parameter δ and simulates the behavior of a literal listener who leverages a literal mapping between article forms and state. The speaker must now compute the posterior probability distributions for the article forms given the state. Importantly, the probability distribution of utterances is inferred based not only on literal interpretations but also on associated costs and α parameter.

At the beginning of the simulation, each agent has an initial cost c for the fully fledged form $c_{} = \log_2 2 = 1$ (the number of words as articulatory targets, e.g. *von dem*, and their logarithm as proposed for example in Gahl et al. (2012)). The application of this cost to the reduced form is null (e.g., *vom*, $c_{<weak>} = \log_2 1 = 0$). However, the reduced form is penalized by a cost accounting for conservativity, which curbs the emergence of novel forms. This cost was operationalized as Shannon’s surprise $c = -\log_2(p)$ (Shannon, 1948) where p is the prior probability of the reduced form.⁵

To sum up, the rational Bayesian speaker updates the probability distribution over forms using the formula proposed in Goodman and Frank (2016) and reported in equation 1.

$$p(u_n|s_n) \propto \alpha(p(s_n|u_n) - c(u_n)) \quad (1)$$

The equation 1 expresses that the probability of a form given a state is proportional to the literal interpretation and the speaker’s rationality, and inversely proportional to its cost.

Once the form u has been chosen by the speaker, the agent with the listener role receives the utterance as input and, using the δ prior returns the probability distribution of the states given the input utterance. The mechanism behind this choice is akin to the method described for the speaker: the prior probability is updated by simulating the behaviour of a pragmatic speaker

⁴One possible interpretation is as follows. Taking the set of contextually salient situations where NP has an extension as U , and the set of situations which contain more than one individual with the NP-property as M , δ' describes the probability of randomly choosing (as a topic) a situation from the complement to M in U , M being almost twice as large in the anaphoric context than otherwise.

⁵We thank Joel Wallenberg for the suggestion to operationalize conservativity this way.

and the listener will guess the state using the posterior distribution.

If the state chosen by the listener corresponds to the one initially assigned to the speaker, the latter receives a reward – modulated by a learning rate parameter $\eta = 1$ – which is added to or subtracted from the cost function, to reward the chosen form and penalize the alternative one. After each game, each agent’s age is augmented by 1 unit. We also introduce a further convolution to simulate some population dynamics that affect language change, such as L1 acquisition. If the agent’s age is greater than their agent’s death parameter, the agent will not participate in any further game and a new agent will be instantiated (so that the number of interacting agents remains constant throughout the simulations). The new agent inherits the cost parameters from a parent and is initially rewarded with a learning rate of 2η to reflect higher sensitivity to primary linguistic data during the L1 acquisition phase, after 12 games, the learning rate is reverted to the default value.

We simulated the referencing game with 40 interacting agents, with each episode featuring pairwise interactions. The simulation ran for 200 episodes (100 in non-anaphoric contexts, 100 in anaphoric contexts). Furthermore, we ran the simulations with two δ values to account for the diffusion of reduced forms through different noun classes: $\delta = 0.9$ (for nouns such as “sun”) and $\delta = 0.5$ (e.g. “dog”). In total, we simulated 8000 games (20 pairs of agents * 200 episodes * 2 NP-classes). For each episode, the mean probability distributions of each form in each state and anaphoric condition were computed and plotted to visually assess the diffusion of the reduced forms. The entire script is downloadable as a Jupyter notebook and detailed code comments: [GitHub repository \(notebook\)](#).

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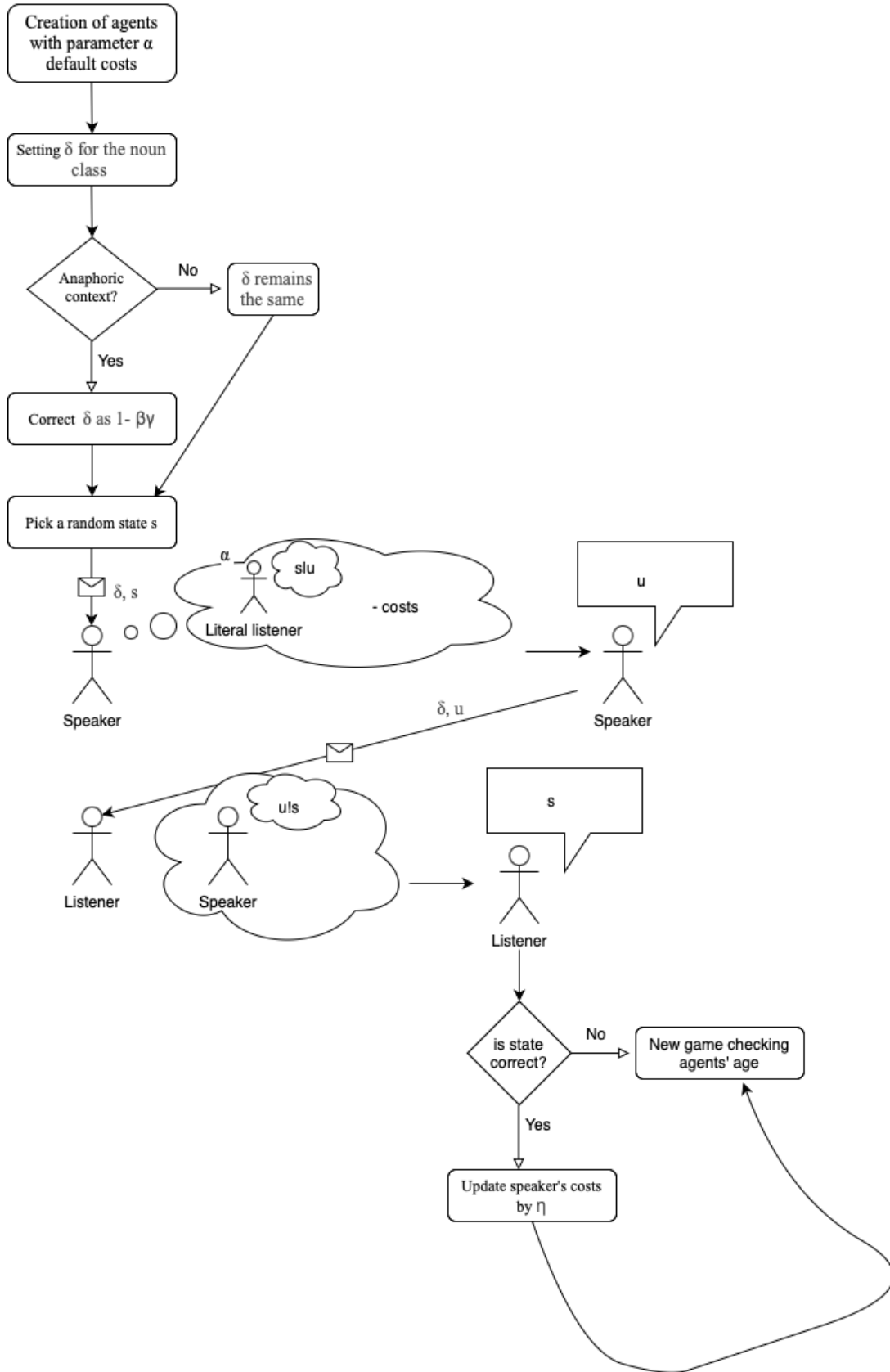


Figure 2: Flowchart of the signaling game

5. Results

We report the results of the simulations in the figures 4 and 3. In each figure, the x-axis indicates the number of episodes in the simulations, serving as an approximation of temporal progression. The y-axis represents the probability (therefore bounded between 0 and 1) that a weak form (e.g. *vom*) is selected given the state $s_2 = \{e_1\}$, i.e. a singleton extension. Note that this probability value is calculated as the mean probability of using the weak form for each agent in that episode. Each plot contains two curves: a red dashed curve showing the temporal evolution of the probability of a weak form in non-anaphoric contexts, and a solid blue curve showing the corresponding probability in anaphoric contexts.

Figure 3 shows both anaphoric and non-anaphoric distributions for the noun class having $\delta = 0.9$ (i.e. "sun", "paradise", "hell" etc.). Here, both curves exhibit a rapid rise already in the early episodes: by the 20th episode, the probability reaches or exceeds 0.70. In later episodes, the curves gradually stabilize around $p \approx 0.9$ –1.0. Overall, a high probability of a singleton extension ($\delta = 0.9$) strongly promotes the use of the weak, reduced form. Notably, the emergence of this form proceeds faster and more decisively in non-anaphoric contexts, whereas anaphoric contexts display a slower convergence and a slightly lower asymptotic probability. Despite this difference, in both cases the reduced form ultimately becomes dominant.

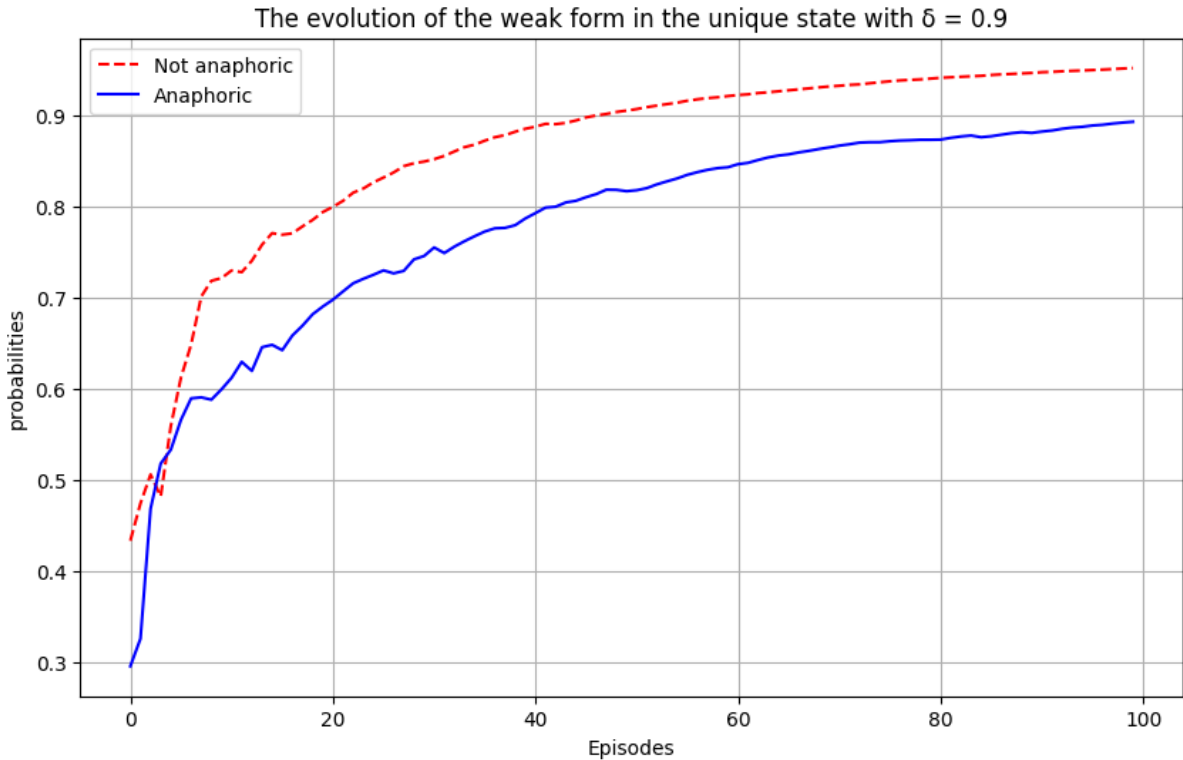


Figure 3: The evolution of the weak form in the unique state, for both anaphoric not-anaphoric contexts with NPs having a $\delta = 0.9$

We now compare the simulations for NPs having a lower $\delta = 0.5$ (e.g. *dog*, *bed*, etc), whose results are reported in figure 4. In this case, the two curves exhibit markedly different trajectories.

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In non-anaphoric contexts, the probability of using the weak form increases steadily over the early episodes and stabilizes around $p \approx 0.7$. In short, we see the same effect as in the case of NPs with $\delta = 0.9$, albeit to a slower degree. The difference in speed between the non-anaphoric curves with different δ values reflects what we have seen in the corpus, where the use of weak forms is initially more frequent with nouns having higher δ .

Going back to the analysis of figure 4, in anaphoric contexts the probability of using a weak form remains consistently low throughout the simulation, never exceeding 0.1 and gradually decreasing after a brief initial fluctuation. Overall, a lower probability of a singleton extension in a random situation ($\delta = 0.5$) substantially limits the emergence of the weak, reduced form, particularly in anaphoric contexts.

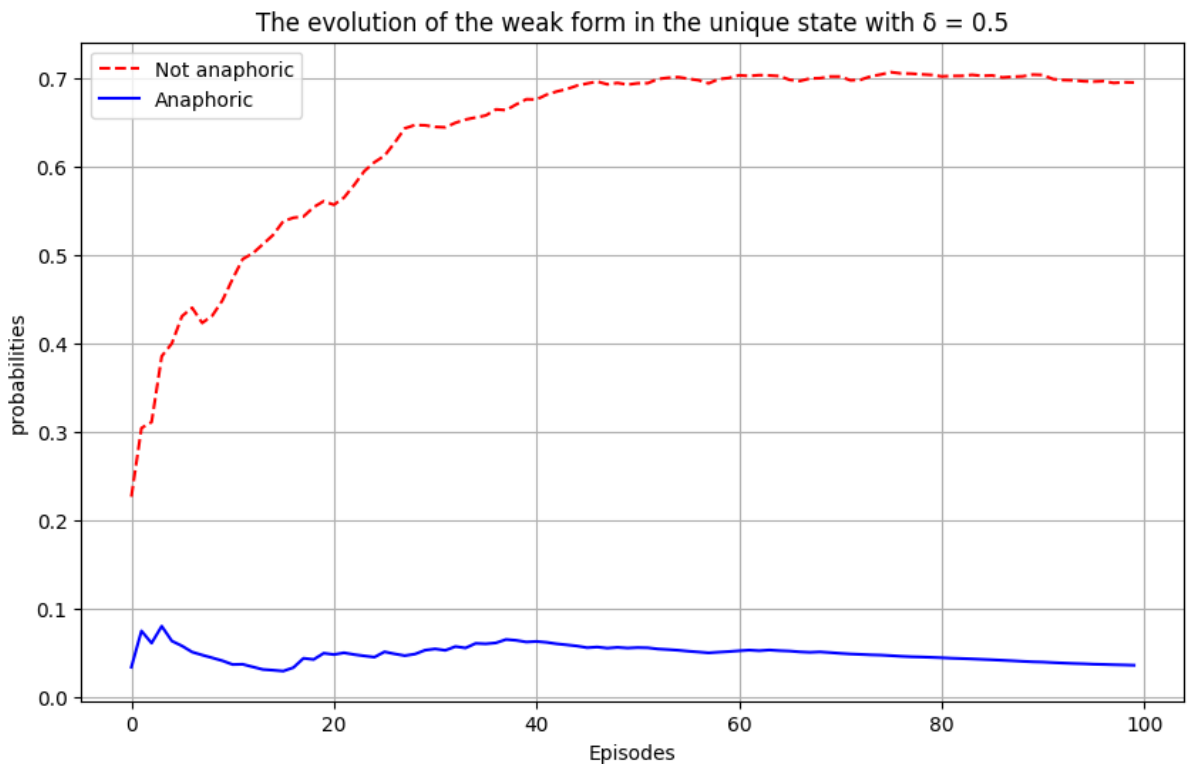


Figure 4: The evolution of the weak form in the unique state, for both anaphoric not-anaphoric contexts with NPs having a $\delta = 0.5$

6. Discussion and conclusion

The goal of this paper was to account for the diachronic emergence and distribution of weak definite articles in German prepositional phrases (e.g. *vom*) at the expense of strong articles (e.g. *von dem*). By combining historical corpus evidence with agent-based simulations grounded in computational pragmatics and reinforcement learning, we aimed to derive this distribution from independently motivated assumptions about reference, economy, and language acquisition. In particular, we assumed that the weak forms, in contrast to strong ones, can give rise to referential uncertainty since their interpretation depends on the properties of the topic situation, as construed by the listener. The uncertainty is low with NPs which are known to mostly denote

singletons and greater with other NP types. We also built into our model an intuition that an indefinite NP lowers the estimates that a contextually salient situation has a singleton with the NP-property in question.

A first robust result of the simulations concerns the behavior of weak forms in non-anaphoric contexts. Here, the weak form spreads rapidly over time. This effect is particularly strong and fast for noun phrases associated with a high probability of having a singleton extension, operationalized in our model as higher values of δ . In such cases, we argue that speakers can safely reduce articulatory effort without incurring a significant risk of referential failure since hearers are highly likely to converge on the intended referent even though the semantics of the weak article itself does not guarantee successful reference. Under these conditions, pressures towards economy favor the use of the reduced form. Through reinforcement learning, repeated communicative success in these contexts strengthens speakers' preference for reduction, ultimately leading to the dominance of the weak form.

This simulation result closely mirrors the earliest stages of the historical development attested in the corpus, where weak articles are typically observed with nouns which highly likely to have a singleton extension in any given situation, such as *paradise*, *hell*, or *sun*. Anaphoric contexts reveal a more complex pattern. In the simulations, the effect of anaphoricity is modulated by the value of δ . For noun phrases with a high δ , weak forms may still emerge under anaphora, although their diffusion is slower than in non-anaphoric contexts. This suggests that even when the lexical properties of NP are such that it is very likely to have a singleton extension, the presence of an indefinite antecedent introduces an additional source of uncertainty that partially counteracts the pressure towards reduction. For noun phrases with lower values of δ , however, the blocking effect of anaphoricity on weak forms is much stronger: they fail to gain ground altogether. Throughout the simulations, their probability remains close to zero, modulo some fluctuations. Crucially, this outcome emerges from the way anaphoricity reshapes prior expectations about reference. The presence of an antecedent increases the likelihood that the current NP may not refer have a singleton extension (in the topic situation), thereby amplifying the uncertainty associated with the use of weak forms. Under these conditions, the communicative cost of reduction outweighs its articulatory benefits in the process of reinforcement learning.

Our model has thus derived the distribution of weak vs. strong forms both in its historical evolution and its modern state. Importantly, the model assumes the same semantics for the D head in both cases, the difference lying in the presence vs. absence of a situation pronoun in the morpho-syntactic structure. The situation pronouns involved in the LF of strong articles must refer to a situation with a singleton extension of a relevant kind, whereas the absence of such pronoun in the LF of weak articles makes their interpretation dependent on whatever happens to be the topic situation, which may or may not contain a singleton of right type. The simulations also contribute to our understanding of how pressures towards clarity and economy jointly shape grammatical systems over time since speakers are assumed to balance the desire to minimize articulatory effort with the need to ensure successful reference.

Beyond the specific case of German definite articles, the present study illustrates the potential of computational pragmatics and agent-based modeling for explaining diachronic change. It may be argued that a similar approach can also be employed in reconstructing the emergence of definite articles, on our account. While they might be initially constrained to anaphoric contexts, they gradually spread onto other contexts (Carlier (2026) and references therein), in a process which appears to a mirror image of the spread of phonologically reduced articles. Such a diffusion could also be framed as the result of a tension between articulatory care and clarity in guiding the hearer toward a correct reference resolution.

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Future work could also extend the model cross-linguistically, exploring whether similar dynamics can account for the distribution of reduced definites in other Germanic varieties. By integrating diachronic data with explicit computational models, we move closer to an explanatory theory of how definiteness systems evolve and stabilize.

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